

Esercizio 1. Let $N \models \mathcal{T}$, ec for $\mathcal{T}^+$, such that $N \models \Phi(a)$. Let $p(x) = pp - tp(a)$ (intended in the new language). Then $\mathcal{T} \cup p(x) \cup \{\exists y \phi(x, y)\}$ is consistent. As a consequence, $N \models \exists y \phi(a, y)$.

Proof. By contradiction, let's say it is inconsistent. Then it is finitely inconsistent. Then $T \models \neg(\exists x \exists y \phi(x, y) \wedge \bigwedge_{k=1, \dots, n} \theta_k(x))$ for some $\theta_1, \dots, \theta_n \in p(x)$. Conjunction of pp formulae is equivalent to a pp formula, so by definition of $\mathcal{T}^+$ we have $\neg(\exists x \Phi(x) \wedge \bigwedge_{k=1, \dots, n} \theta_k(x)) \in \mathcal{T}^+$. Then $N \models p(a) \wedge \Phi(a) \wedge \neg(\exists x \Phi(x) \wedge \bigwedge_{k=1, \dots, n} \theta_k(x))$, a contradiction.

By compactness, we can find a model M of $\mathcal{T}$ such that for some $c \in M^x$ we have $M \models p(c) \wedge \exists y \phi(c, y)$. Let M' be an $|N|^+$ -saturated elementary extension of M . Then one can extend $a \mapsto c$ to a morphism from $N \rightarrow M'$. Since $M' \models \exists y \phi(c, y)$ and N is ec (in the new language, therefore in the original) we get $N \models \exists y \phi(a, y)$. □

Esercizio 2. Let N ec model for $\mathcal{T}^+$ and assume $N \models \exists y \phi(a, y)$. Then $N \models \Phi(a)$.

Proof. By contradiction, say $N \not\models \Phi(a)$. Now consider N' , whose semantic is the same as N with the only difference that $N' \models \Phi(a)$. We claim that $N' \models \mathcal{T}^+$. In fact otherwise there would be some sentence of the form $\rho \equiv \exists x_1, \dots, x_l \bigwedge_{k=1, \dots, n} Q_k(\vec{x})$ such that $N' \models \rho$ but $\mathcal{T} \models \neg \rho'$, where ρ' is obtained as the substitution of every occurrence in ρ of Φ with $\exists y \phi$. Let $\vec{b}$ be the solution of ρ in N' and let $\vec{b}_k$ be what occurs in Q_k . Then split the istantion of ρ as:

$$\bigwedge_{k: Q_k \neq \Phi} Q_k(b_k) \wedge \bigwedge_{k: Q_k = \Phi, \vec{b}_k = \vec{a}} Q_k(a) \wedge \bigwedge_{k: Q_k = \Phi, \vec{b}_k \neq \vec{a}} Q_k(\vec{b}_k)$$

Now observe that

$$N \models \bigwedge_{k: Q_k \neq \Phi} Q_k(b_k) \wedge \bigwedge_{k: Q_k = \Phi, \vec{b}_k = \vec{a}} \exists y \phi(a, y) \wedge \bigwedge_{k: Q_k = \Phi, \vec{b}_k \neq \vec{a}} Q_k(\vec{b}_k)$$

In fact:

1. the first and third $\bigwedge$ are ok because N, N' are the same here and N' models it.
2. the second is ok because by hypothesis $N \models \exists y \phi(a, y)$.

Finally, by the previous exercise we get

$$N \models \bigwedge_{k: Q_k \neq \Phi} Q_k(b_k) \wedge \bigwedge_{k: Q_k = \Phi, \vec{b}_k = \vec{a}} \exists y \phi(a, y) \wedge \bigwedge_{k: Q_k = \Phi, \vec{b}_k \neq \vec{a}} \exists y \phi(\vec{b}_k, y)$$

But this is a contradiction since this means that $N \models \rho$, so $N \not\models \mathcal{T}^+$. So far we proved that $N' \models \mathcal{T}^+$. Now observe that $N \rightarrow N', n \mapsto n$ is a morphism and $N' \models \Phi(a)$. Then using N ec we get $N \models \Phi(a)$. □